
1 The language of synthetic category theory

At the heart of mathematics lies the concept of a set, a collection of mathematical objects. In modern mathematics, sets serve as the building blocks for defining everything from numbers and relations to more complex structures like groups, rings, and topological spaces. Sets are manipulated using basic operations like forming unions, intersections, and complements, and are compared using functions between sets. It is fair to say that the language of set theory forms an essential toolkit for mathematicians.

Although formal axiomatizations of set theory exist, such as those by Zermelo and Fraenkel, they often diverge from the working mathematician's intuitive understanding of sets. For instance, in formal set theory, every natural number needs to be defined as a set, and in particular one is allowed to speak of the elements of a natural number. Similarly, every function between sets needs to be defined as a set. While this approach is formally valid, it can seem disconnected from everyday mathematical discourse.

In practice, mathematicians often use a more pragmatic approach known as *naive set theory*. In naive set theory, one works with sets based on intuitive understanding, without invoking formal logical or axiomatic implementations. For example, the set of natural numbers is simply the set containing the numbers 0, 1, 2, etcetera, and functions between sets X and Y are simply 'specifications' that assign to every element of X an element of Y . The simplicity and accessibility of naive set theory make it relatively easy for students to learn it, and indeed in many introductory mathematics textbooks sets are used in the naive way without being rigorously introduced.

A similar trend is occurring in the field of higher category theory. In recent decades, higher category theory has seen rapid growth, in large part thanks to contributions from people like Joyal, Lurie, and others. Their work provides solid foundations for the field, and the language of higher categories is now extending beyond its origins in algebraic topology to fields like algebraic geometry and representation theory. But while the original works of Joyal and Lurie are based on explicit set-theoretical models for higher categories called 'quasicategories', more and more research papers are written in a 'model-independent' way in which the usage of details about the specific choice of implementations is avoided.

We believe that a model-independent language makes higher category theory more accessible, just as naive set theory makes it possible to learn and use sets without first developing formal set theory. With this in mind, the goal of this book is to provide a detailed treatment of the foundations of higher category theory in the language of *synthetic category theory*, without reference to specific implementations.

The goal of the present chapter is to introduce this language and specify the rules governing its use. We introduce the basic notions, the operations on them, and the identifications relating these operations. Alongside these rules, we develop term notation for displaying functors and natural isomorphisms as expressions. In particular, our approach is *synthetic* in nature rather than *analytic*: constructions of categories, like products or functor categories, are not *defined* in terms of set-level constructions, but are *axiomatized* to exist and satisfy the expected properties.

1.1 The basic vocabulary

We start by introducing the basic vocabulary of synthetic category theory, taking inspiration from naive set theory. In the same way that functions between sets are taken as a primitive notion in naive set theory, as opposed to being defined via their graph, we similarly take functors between categories as a primitive notion, together with the basic relations they should satisfy.

In synthetic category theory, the role of collections is played by *animae* rather than sets. The principal distinction between sets and animae is that an identification between objects of an anima is structure rather than a mere property. Such identifications should therefore themselves form a collection of mathematical objects, hence an anima.

We initially treat animae as a primitive distinguished class of categories, leaving their structural characterization among categories unspecified. In the next chapter, we identify the groupoids among all categories and demand that a category is an anima precisely when it is a groupoid.

Epistemic status: The choice of foundational language introduced here is still evolving, and has become more and more refined over time. It was obtained by starting with the language of ‘naive category theory’ used by modern homotopy theorists, and refining it over time by adding more coherences whenever the need became transparent. We expect the current list of axiomatized coherences to be still incomplete, and our natural language treatment should not be thought of as an attempt of a full formalization of synthetic category theory.

1.1.1 Categories, functors and natural isomorphisms

Axiom A.1. The following are the primitives in the vocabulary of synthetic category theory:

- (1) There are *categories*, which we denote by symbols like C , D , and E . Among categories, there are *animae*; these encode the idea of a homotopy-coherent collection.
- (2) For categories C and D , there is a primitive notion of a *functor* $f: C \rightarrow D$. Every category has an *identity functor* $\text{id}_C: C \rightarrow C$. Given $f: C \rightarrow D$ and $g: D \rightarrow E$, there is a *composite functor* $g \circ f: C \rightarrow E$.
- (3) Given parallel functors $f, g: C \rightarrow D$, there is an anima $(f \cong g)$. For $h: D \rightarrow E$ and $k: B \rightarrow C$, there are functors

$$h \star -: (f \cong g) \rightarrow (hf \cong hg), \quad - \star k: (f \cong g) \rightarrow (fk \cong gk),$$

called *postwhiskering* and *prewhiskering*, respectively. There is also an *inverse functor*

$$(-)^{-1} : (f \cong g) \rightarrow (g \cong f).$$

Definition 1.1.1 (Terms). A functor $f : C \rightarrow D$ may be displayed in *term notation* as

$$x : C \vdash f(x) : D.$$

The functor and the expression $f(x)$ are two notations for the same datum. We call $x : C$ its *context*. More generally, we may write $C \vdash t : D$ when the name of the variable is left implicit.

Composition of functors is displayed by substitution: for $g : D \rightarrow E$, the composite $g \circ f$ is written $g(f(x))$. Thus an expression with values in D and variable $x : C$ always represents a functor $C \rightarrow D$.

1.1.2 The terminal category and products

We introduce a distinguished anima $*$, which will be the terminal category. Before stating its universal property, let us use it to define absolute objects and natural isomorphisms.

Definition 1.1.2 (Absolute object). An *absolute object* $c : C$ of a category C is a functor $c : * \rightarrow C$. We sometimes also write $c \in C$. An *isomorphism* between absolute objects $c, c' : C$ is a natural isomorphism between the functors $c, c' : * \rightarrow C$.

Definition 1.1.3. Given functors $f, g : C \rightarrow D$, a *natural isomorphism* $\alpha : f \cong g$ is an absolute object of $(f \cong g)$, i.e. a functor $\alpha : * \rightarrow (f \cong g)$.

Two natural isomorphisms are themselves parallel functors, so that their identifications again form an anima. The same construction may be iterated. We may display α in term notation as

$$x : C \vdash \alpha(x) : f(x) \cong g(x).$$

This means a natural isomorphism of the two functors f and g . The same convention applies to identifications between natural isomorphisms. The notation $\cong$ for identifications is used to evoke the idea of isomorphisms in category theory. The following principle guides our use of this language:

Fundamental principle of homotopy theory: Whenever we would like to express that two things *are equal*, we should remember *how* they are equal by specifying an identification between them.

Definition 1.1.4 (Equivalence of categories). A functor $f : C \rightarrow D$ is said to be an *equivalence* if there exists another functor $g : D \rightarrow C$ equipped with natural isomorphisms $\text{id}_C \cong g f$ and $\text{id}_D \cong f g$. We refer to g as the *inverse* of f . Equivalences will often be introduced by writing $f : C \xrightarrow{\sim} D$. We may write $C \simeq D$ to indicate that C and D are equivalent via an unnamed equivalence.

Axiom A.2 (Terminal category). There is an anima $*$, called the *terminal category*. For every category T , there is a functor $p_T: T \rightarrow *$. For any functors $f, g: T \rightarrow *$, the functor

$$p_{(f \cong g)}: (f \cong g) \longrightarrow *$$

is an equivalence.

In particular, any two functors to $*$ are naturally isomorphic, and the anima of such natural isomorphisms is contractible.

Definition 1.1.5 (Constant functor). Let C and D be categories and let x be an absolute object of D . We define the *constant functor* $\text{const}_x: C \rightarrow D$ as the composite $x \circ p_C: C \rightarrow D$:

$$\text{const}_x: C \xrightarrow{p_C} * \xrightarrow{x} D.$$

Definition 1.1.6 (Contractible category). A category C is called *contractible* if the functor $p_C: C \rightarrow *$ is an equivalence.

Remark 1.1.7. Observe that a category C is contractible if and only if there exists an absolute object $x: * \rightarrow C$ such that the composite $\text{const}_x: C \rightarrow C$ is naturally isomorphic to the identity $\text{id}_C: C \rightarrow C$. A natural isomorphism $H: \text{const}_x \cong \text{id}_C$ is called a *contraction of C onto x* .

Products of categories

Next we introduce the *product* of two categories.

Axiom A.3 (Product of categories). The *product* of categories C and D is a category $C \times D$ equipped with *projection functors* $\text{pr}_1: C \times D \rightarrow C$ and $\text{pr}_2: C \times D \rightarrow D$.

Given $f: T \rightarrow C$ and $g: T \rightarrow D$, there is a functor $(f, g): T \rightarrow C \times D$ equipped with natural isomorphisms $\text{pr}_1(f, g) \cong f$ and $\text{pr}_2(f, g) \cong g$. For any $h, k: T \rightarrow C \times D$, the functor

$$(\text{pr}_1 \star -, \text{pr}_2 \star -): (h \cong k) \longrightarrow (\text{pr}_1 h \cong \text{pr}_1 k) \times (\text{pr}_2 h \cong \text{pr}_2 k) \quad (1.1)$$

is an equivalence. If C and D are animae, then $C \times D$ is an anima.

In term notation, the pair is written $(f(x), g(x))$, with identifications $\text{pr}_1(f(x), g(x)) \cong f(x)$ and $\text{pr}_2(f(x), g(x)) \cong g(x)$. For arbitrary expressions $h(x), k(x): C \times D$, natural isomorphisms $\alpha(x): \text{pr}_1 h(x) \cong \text{pr}_1 k(x)$ and $\beta(x): \text{pr}_2 h(x) \cong \text{pr}_2 k(x)$ give a natural isomorphism $(\alpha(x), \beta(x)): h(x) \cong k(x)$ by applying an inverse of the displayed equivalence. Its two projections are identified with α and β .

Convention 1.1.8. We may abbreviate a single variable $z: C \times D$ by $x: C, y: D$ when the expression under consideration uses z only through $\text{pr}_1(z)$ and $\text{pr}_2(z)$. In this display, x stands for $\text{pr}_1(z)$ and y for $\text{pr}_2(z)$. For more than two variables we use a fixed parenthesization of their product.

1.1.3 Coherences

In order for composition to behave as expected, we specify its units and associativity isomorphisms and their compatibility with natural isomorphisms.

Axiom A.4. For every functor f , there is an identity natural isomorphism $\text{id}_f: f \cong f$. For parallel functors $f, g, h: C \rightarrow D$, there is a composition functor

$$- \circ -: (g \cong h) \times (f \cong g) \longrightarrow (f \cong h).$$

This composition is unital and associative, and the inverse functor supplies two-sided inverses, through specified natural isomorphisms of functors on the corresponding isomorphism animae. On objects these laws read

$$\text{id}_g \circ \alpha \cong \alpha \cong \alpha \circ \text{id}_f, \quad (\gamma \circ \beta) \circ \alpha \cong \gamma \circ (\beta \circ \alpha),$$

$$\alpha^{-1} \circ \alpha \cong \text{id}_f, \quad \alpha \circ \alpha^{-1} \cong \text{id}_g.$$

For functors $f: C \rightarrow D$, $g: D \rightarrow E$ and $h: E \rightarrow F$, there are natural isomorphisms

$$\lambda_f: \text{id}_D \circ f \cong f, \quad \rho_f: f \circ \text{id}_C \cong f, \quad \alpha_{f,g,h}: (h \circ g) \circ f \cong h \circ (g \circ f).$$

Axiom A.5. Prewhiskering and postwhiskering preserve identity natural isomorphisms and vertical composition through specified natural isomorphisms of functors on the input isomorphism animae. For functors $f, g, h: C \rightarrow D$, $k: B \rightarrow C$ and $u: D \rightarrow E$, these laws read

$$u \star \text{id}_f \cong \text{id}_{u \circ f}, \quad \text{id}_f \star k \cong \text{id}_{f \circ k},$$

$$u \star (\tau \circ \sigma) \cong (u \star \tau) \circ (u \star \sigma), \quad (\tau \circ \sigma) \star k \cong (\tau \star k) \circ (\sigma \star k),$$

where $\sigma: f \cong g$ and $\tau: g \cong h$. Whiskering by an identity functor, by a composite, and in the two possible orders is compared using the unitors and associator of functor composition. For $\sigma: f \cong g$, the identity laws are

$$\lambda_g \circ (\text{id}_D \star \sigma) \cong \sigma \circ \lambda_f, \quad \rho_g \circ (\sigma \star \text{id}_C) \cong \sigma \circ \rho_f.$$

For additional functors $l: A \rightarrow B$ and $v: E \rightarrow F$, the composite laws are

$$\alpha_{g,u,v} \circ ((v \circ u) \star \sigma) \cong (v \star (u \star \sigma)) \circ \alpha_{f,u,v},$$

$$\alpha_{l,k,g} \circ ((\sigma \star k) \star l) \cong (\sigma \star (k \circ l)) \circ \alpha_{l,k,f}.$$

Finally, the two orders of whiskering are related by

$$\alpha_{k,g,u} \circ ((u \star \sigma) \star k) \cong (u \star (\sigma \star k)) \circ \alpha_{k,f,u}.$$

The last five comparisons are natural isomorphisms of functors on $(f \cong g)$, and the two vertical-composition comparisons are natural isomorphisms of functors on $(g \cong h) \times (f \cong g)$. The two ways of pasting natural isomorphisms are identified. More precisely, for functors $F, G: D \rightarrow E$ and $h, k: T \rightarrow D$, there is a specified natural isomorphism between the functors $(F \cong G) \times (h \cong k) \rightarrow (Fh \cong Gk)$

$$(\tau \star k) \circ (F \star \sigma) \cong (G \star \sigma) \circ (\tau \star h), \quad (1.2)$$

where τ and σ are the two projection functors from $(F \cong G) \times (h \cong k)$. Fixing either input gives the corresponding comparison by restriction, using that input as a constant functor.

Construction 1.1.9 (Horizontal composition). Given $\alpha: f \cong f'$ and $\beta: g \cong g'$, we write

$$\beta * \alpha := (\beta \star f') \circ (g \star \alpha) : gf \cong g'f'$$

for their *horizontal composite*. Interchange compares this with $(g' \star \alpha) \circ (\beta \star f)$. Its action on identifications is obtained from the functors just introduced.

Lemma 1.1.10. *Horizontal composition preserves identities and vertical composites through specified natural isomorphisms on the products of the input isomorphism animae.*

Proof. Apply the whiskering laws and joint interchange Equation (1.2). □

Definition 1.1.11. A *commutative square of functors*, often displayed by means of a diagram

$$\begin{array}{ccc} C & \xrightarrow{f} & D \\ g \downarrow & & \downarrow h \\ C' & \xrightarrow{f'} & D', \end{array}$$

consists of four functors f, g, h and f' with sources and targets as displayed in the diagram, together with a specified natural isomorphism $\alpha: h \circ f \cong f' \circ g$. We may similarly define a *commutative square of natural isomorphisms*, etcetera.

We will often refer to the data of just the four functors f, g, h and f' as a *square*, and say that *the square commutes* whenever a natural isomorphism $\alpha: h \circ f \cong f' \circ g$ can be provided.

A *commutative triangle of functors*, displayed by means of a diagram

$$\begin{array}{ccc} & D & \\ f \nearrow & & \searrow g \\ C & \xrightarrow{h} & E, \end{array}$$

consists of three functors f, g and h with sources and targets as displayed in the diagram, together with a specified natural isomorphism $\alpha: h \cong g \circ f$. We may similarly define a *commutative triangle of natural isomorphisms*, etcetera.

We will similarly display more general diagrams of functors by drawing various directed planar graphs with vertices labeled by categories and with edges labeled by functors. We say the diagram

commutes if every face comes equipped with a natural isomorphism between the two composites of functors that can be formed along the boundary of that face.

Axiom A.6. Horizontal composition of natural isomorphisms is compatible with the unitality and associativity isomorphisms from Axiom A.4:

- For a natural isomorphism $\alpha: f \cong f'$ of functors $C \rightarrow D$ there are commutative squares of natural isomorphisms

$$\begin{array}{ccc} \text{id}_D \circ f & \xrightarrow[\cong]{\text{id}_D * \alpha} & \text{id}_D \circ f' \\ \lambda_f \downarrow & & \downarrow \lambda_{f'} \\ f & \xrightarrow[\cong]{\alpha} & f' \end{array} \quad \text{and} \quad \begin{array}{ccc} f \circ \text{id}_C & \xrightarrow[\cong]{\alpha * \text{id}_C} & f' \circ \text{id}_C \\ \rho_f \downarrow & & \downarrow \rho_{f'} \\ f & \xrightarrow[\cong]{\alpha} & f'. \end{array}$$

- For natural isomorphisms $\alpha: f \cong f'$, $\beta: g \cong g'$ and $\gamma: h \cong h'$, the following square of natural isomorphisms commutes:

$$\begin{array}{ccc} (h \circ g) \circ f & \xrightarrow[\cong]{(\gamma * \beta) * \alpha} & (h' \circ g') \circ f' \\ \alpha_{f,g,h} \downarrow & & \downarrow \alpha_{f',g',h'} \\ h \circ (g \circ f) & \xrightarrow[\cong]{\gamma * (\beta * \alpha)} & h' \circ (g' \circ f'). \end{array}$$

These comparisons are natural isomorphisms of the corresponding functors on products of the input isomorphism animae. Their restrictions along functors into those animae give the same formulas for identifications.

Remark 1.1.12. It will sometimes be convenient to abbreviate the conditions from Axiom A.6 as follows:

$$\text{id} * \alpha \cong \alpha \cong \alpha * \text{id} \quad \text{and} \quad (\gamma * \beta) * \alpha \cong \gamma * (\beta * \alpha).$$

More generally, given natural isomorphisms $\alpha: f \cong g$ and $\alpha': f' \cong g'$ of functors $C \rightarrow D$, together with specified endpoint comparisons $s: f \cong f'$ and $t: g \cong g'$, we write $\alpha \cong \alpha'$ for an identification $t \circ \alpha \cong \alpha' \circ s$, displayed as a commutative square:

$$\begin{array}{ccc} f & \xrightarrow[\cong]{\alpha} & g \\ s \downarrow & & \downarrow t \\ f' & \xrightarrow[\cong]{\alpha'} & g'. \end{array}$$

These endpoint comparisons are kept fixed when comparing pastings.

Axiom A.7 (Pentagon and triangle identities). For functors $f: A \rightarrow B$, $g: B \rightarrow C$, $h: C \rightarrow D$ and $k: D \rightarrow E$, the following pentagon of natural isomorphisms commutes:

$$\begin{array}{ccc}
 & (k \circ h) \circ (g \circ f) & \\
 \alpha_{f,g,k \circ h} \nearrow & & \searrow \alpha_{g \circ f, h, k} \\
 ((k \circ h) \circ g) \circ f & & k \circ (h \circ (g \circ f)) \\
 \alpha_{g,h,k} * \text{id}_f \downarrow & & \uparrow \text{id}_k * \alpha_{f,g,h} \\
 (k \circ (h \circ g)) \circ f & \xrightarrow{\alpha_{f,h \circ g, k}} & k \circ ((h \circ g) \circ f)
 \end{array}$$

Moreover, for composable functors $f: C \rightarrow D$ and $g: D \rightarrow E$, the following triangle of natural isomorphisms commutes:

$$\begin{array}{ccc}
 (g \circ \text{id}_D) \circ f & \xrightarrow{\alpha_{f, \text{id}_D, g}} & g \circ (\text{id}_D \circ f) \\
 \rho_g * \text{id}_f \searrow & & \downarrow \text{id}_g * \lambda_f \\
 & & g \circ f.
 \end{array}$$

1.1.4 Terms

Let us collect some conventions for the term notation introduced above. An expression $x: C \vdash f(x): D$ represents a functor $C \rightarrow D$; the category D is fixed. For a functor $\sigma: B \rightarrow C$, substitution gives the expression $y: B \vdash f(\sigma(y)): D$, representing $f \circ \sigma$. The unitality and associativity comparisons for substitution are those of composition of functors.

A natural isomorphism of expressions means a natural isomorphism of their whole functors. For example, $x: C \vdash \alpha(x): f(x) \cong g(x)$ is a point of $(f \cong g)$, where $f, g: C \rightarrow D$. A functor from another category to this fixed isomorphism anima is a different expression, with that anima as its target.

Definition 1.1.13 (Universal term). The identity functor $\text{id}_C: C \rightarrow C$ is displayed as $x: C \vdash x: C$. We call this the *universal term* of C .

An already constructed functor $\Phi: C \rightarrow D$ acts on terms by composition. Natural isomorphisms between such functors act by prewhiskering. As further constructions become available, we use the same convention to display their functors and natural isomorphisms as operations on expressions.

The notation is inspired by type-theoretic approaches to synthetic category theory, such as [RS17; GWB24]. Here substitution is composition of functors, and its laws are the specified isomorphisms. A functor and its term display are simply two representations of the same datum.

1.2 Equivalences of categories

We now develop some basic properties of the equivalences introduced in Definition 1.1.4.

1.2.1 Equivalences and lifting

Proposition 1.2.1. *Let $f: C \rightarrow D$ be a functor. For f to be an equivalence, it suffices that the following properties hold for every category T :*

- (1) *For $a, b: T \rightarrow C$, any natural isomorphism $fa \cong fb$ gives a natural isomorphism $a \cong b$.*
- (2) *For any $d: T \rightarrow D$, there is a functor $c: T \rightarrow C$ equipped with a natural isomorphism $fc \cong d$.*

Proof. Apply the second property to id_D to obtain $g: D \rightarrow C$ and $fg \cong \text{id}_D$. The associator, this isomorphism, and the unitors identify $f(gf)$ with $f\text{id}_C$. The first property, with $T = C$, gives $gf \cong \text{id}_C$. $\square$

Definition 1.2.2 (Section/retraction). Consider a functor $f: C \rightarrow D$. A *section* of f is a functor $s: D \rightarrow C$ equipped with a natural isomorphism $fs \cong \text{id}_D$. A *retraction* of f is a functor $r: D \rightarrow C$ equipped with a natural isomorphism $\text{id}_C \cong rf$.

Lemma 1.2.3. *A functor $f: C \rightarrow D$ is an equivalence if and only if it admits both a section s and a retraction r . In this case there exists a natural isomorphism $s \cong r$ and s is an inverse of f .*

Proof. If f is an equivalence with inverse $g: D \rightarrow C$, then clearly g is both a section and a retraction of f . Conversely, given a section $s: D \rightarrow C$, $fs \cong \text{id}_D$ and a retraction $r: D \rightarrow C$, $\text{id}_C \cong rf$, we obtain a natural isomorphism $s \cong r$ by composing the following chain of natural isomorphisms:

$$s \cong \text{id}_C \circ s \cong (r \circ f) \circ s \cong r \circ (f \circ s) \cong r \circ \text{id}_D \cong r.$$

It follows that f is an equivalence with inverse s , since there are natural isomorphisms $fs \cong \text{id}_D$ and $sf \cong rf \cong \text{id}_C$. $\square$

Corollary 1.2.4. *Any two inverses g and h of an equivalence f are isomorphic.* $\square$

Lemma 1.2.5. *If $f: C \rightarrow D$ and $f': D \rightarrow E$ are equivalences with inverses $g: D \rightarrow C$ and $g': E \rightarrow D$, then also the composite $f'f: C \rightarrow E$ is an equivalence with inverse $gg': E \rightarrow C$.*

Proof. We have $gg'f'f \cong g\text{id}_D f \cong gf \cong \text{id}_C$ and $f'f gg' \cong f'\text{id}_D g' \cong f'g' \cong \text{id}_E$. $\square$

Lemma 1.2.6. *A functor naturally isomorphic to an equivalence is an equivalence.*

Proof. The same inverse works, by whiskering with the given natural isomorphism. $\square$

Lemma 1.2.7. *Equivalences of categories satisfy the 2-out-of-3 property: given functors $f: C \rightarrow D$ and $g: D \rightarrow E$, if two of the three functors f , g and gf are equivalences, then so is the third.*

Proof. If g and f are equivalences, then so is gf by Lemma 1.2.5. If f and gf are equivalences, then we may rewrite g as the composite $g \cong g \circ \text{id}_D \cong g \circ f \circ f^{-1} \cong gf \circ f^{-1}$, so that g is the composite of two equivalences and hence itself an equivalence. A similar argument shows that f is an equivalence whenever g and gf are equivalences. $\square$

Lemma 1.2.8. *Equivalences of categories satisfy the 2-out-of-6 property: given functors $f: C \rightarrow D$, $g: D \rightarrow E$ and $h: E \rightarrow F$ such that gf and hg are equivalences, also the functors f , g , h and hgf are equivalences.*

Proof. Consider the functors $f' := f \circ (gf)^{-1} : E \rightarrow D$ and $h' := (hg)^{-1} \circ h : E \rightarrow D$. Then there are natural isomorphisms

$$g \circ f' \cong g \circ f \circ (gf)^{-1} \cong gf \circ (gf)^{-1} \cong \text{id}_E$$

and

$$h' \circ g \cong (hg)^{-1} \circ h \circ g \cong (hg)^{-1} \circ hg \cong \text{id}_D.$$

Hence it follows from Lemma 1.2.3 that g is an equivalence. In light of Lemma 1.2.7, we then get that also f and h are equivalences, and then so is hg . $\square$

Definition 1.2.9. We say that a functor $f : C \rightarrow D$ is a *retract* of a functor $f' : C' \rightarrow D'$ if there exists a commutative diagram of the form

$$\begin{array}{ccccc} C & \xrightarrow{g} & C' & \xrightarrow{h} & C \\ f \downarrow & & \downarrow f' & & \downarrow f \\ D & \xrightarrow{k} & D' & \xrightarrow{j} & D, \end{array}$$

together with natural isomorphisms $\alpha : \text{id}_C \cong h \circ g$ and $\beta : j \circ k \cong \text{id}_D$, and an isomorphism between $\text{id}_f : f \cong f$ and the composite natural isomorphism

$$f \cong f \circ \text{id}_C \xrightarrow{\alpha} f \circ h \circ g \cong j \circ f' \circ g \cong j \circ k \circ f \xrightarrow{\beta} \text{id}_D \circ f \cong f.$$

Lemma 1.2.10. Assume that $f : C \rightarrow D$ is a retract of $f' : C' \rightarrow D'$. If f' is an equivalence, then so is f .

Proof. Let $(f')^{-1} : D' \rightarrow C'$ be an inverse of f' . We claim that the composite $w := h \circ (f')^{-1} \circ k : D \rightarrow C$ is inverse to f . Indeed, we have natural isomorphisms

$$f \circ w \cong f \circ h \circ (f')^{-1} \circ k \cong j \circ f' \circ (f')^{-1} \circ k \cong j \circ k \cong \text{id}_D$$

and

$$w \circ f \cong h \circ (f')^{-1} \circ k \circ f \cong h \circ (f')^{-1} \circ f' \circ g \cong h \circ g \cong \text{id}_C.$$

This finishes the proof. $\square$

Lemma 1.2.11. *Prewiskering or postwhiskering by an equivalence induces equivalences of isomorphism animae.*

Proof. Composition with a fixed natural isomorphism is an equivalence, with inverse given by composition with its inverse. This follows from the vertical unit, associativity, and inverse laws.

Let $F : C \rightarrow D$ have inverse G , and let $a, b : T \rightarrow C$. Consider

$$(a \cong b) \xrightarrow{F\star-} (Fa \cong Fb) \xrightarrow{G\star-} (GFa \cong GFb) \xrightarrow{F\star-} (FGFa \cong FGfb).$$

Interchange with $GF \cong \text{id}_C$ identifies the composite of the first two functors with transport along its endpoint isomorphisms. Hence that composite is an equivalence. The composite of the last two is an equivalence for the same reason, using $FG \cong \text{id}_D$. The 2-out-of-6 property proves the claim. Prewiskering is treated in the same way. $\square$

Corollary 1.2.12 (Lifting through equivalences). *Prewhiskering or postwhiskering by an equivalence lifts a specified natural isomorphism, together with an identification of its image with that isomorphism. It also lifts and reflects identifications between natural isomorphisms.*

Proof. Apply an inverse of the equivalence of isomorphism animae from Lemma 1.2.11, and apply the same lemma to that equivalence for the assertion about identifications. $\square$

1.2.2 Products and their comparisons

Remark 1.2.13. Given categories C and D , a *span* from C to D is a triple (T, f, g) consisting of functors $f: T \rightarrow C$ and $g: T \rightarrow D$. Axiom A.3 makes $(C \times D, \text{pr}_1, \text{pr}_2)$ the *terminal span*: there is a map $(f, g): T \rightarrow C \times D$ with specified comparisons on both legs, and any two such maps are naturally isomorphic compatibly with those comparisons. Indeed, lift the two leg comparisons through Equation (1.1); its higher lifting property supplies the corresponding identifications between comparisons.

From the universal property of products, we may directly deduce that products are commutative, unital, and associative. For commutativity of products, one argues as follows:

Lemma 1.2.14. *For categories C and D , the functor $(\text{pr}_2, \text{pr}_1): C \times D \rightarrow D \times C$ is an equivalence, with inverse given by $(\text{pr}_2, \text{pr}_1): D \times C \rightarrow C \times D$.*

Proof. We need to provide natural isomorphisms $(\text{pr}_2, \text{pr}_1) \circ (\text{pr}_2, \text{pr}_1) \cong \text{id}_{C \times D}$ and $(\text{pr}_2, \text{pr}_1) \circ (\text{pr}_2, \text{pr}_1) \cong \text{id}_{D \times C}$. By symmetry, it will suffice to produce the first natural isomorphism. For this, it will in turn suffice to produce natural isomorphisms after composing with the functors pr_1 and pr_2 . For the composition with pr_1 we compute that

$$\text{pr}_1 \circ ((\text{pr}_2, \text{pr}_1) \circ (\text{pr}_2, \text{pr}_1)) \cong (\text{pr}_1 \circ (\text{pr}_2, \text{pr}_1)) \circ (\text{pr}_2, \text{pr}_1) \cong \text{pr}_2 \circ (\text{pr}_2, \text{pr}_1) \cong \text{pr}_1,$$

which is indeed isomorphic to $\text{pr}_1 \circ \text{id}_{C \times D}$. The case for composition with pr_2 is analogous. This finishes the proof. $\square$

Unitality may be argued similarly:

Lemma 1.2.15. *For every category C , the projection functor $\text{pr}_1: C \times * \rightarrow C$ is an equivalence, with inverse given by the functor $(\text{id}_C, p_C): C \rightarrow C \times *$.*

Proof. The composite $\text{pr}_1 \circ (\text{id}_C, p_C): C \rightarrow C$ is isomorphic to id_C by the defining property of (id_C, p_C) . To show that the composite $(\text{id}_C, p_C) \circ \text{pr}_1: C \times * \rightarrow C \times *$ is isomorphic to $\text{id}_{C \times *}$, it will suffice to do so after composing with pr_1 and with pr_2 . The case for pr_2 is clear, since any two functors $C \times * \rightarrow *$ are isomorphic. In the case of pr_1 we compute

$$\text{pr}_1 \circ ((\text{id}_C, p_C) \circ \text{pr}_1) \cong (\text{pr}_1 \circ (\text{id}_C, p_C)) \circ \text{pr}_1 \cong \text{id}_C \circ \text{pr}_1 \cong \text{pr}_1 \cong \text{pr}_1 \circ \text{id}_{C \times *},$$

where the first isomorphism is associativity, the second is the relation proved in the first paragraph, and the third and fourth are unitality. This finishes the proof. $\square$

Remark 1.2.16. In the previous proofs, we have been very explicit about the usage of associativity and naturality isomorphisms. In the remainder of this book, we will often leave the usage of such isomorphisms implicit in order to make the proofs more concise. Comparisons between pastings use the specified routes, with their endpoint comparisons retained.

Exercise 1.2.17 (Exercise 1.12.1 (outside selection)). Formulate and prove the associativity of the product of categories.

Construction 1.2.18. The formation of products of categories is functorial: given two functors $f: C \rightarrow C'$ and $g: D \rightarrow D'$, we obtain a new functor

$$f \times g := (f \circ \text{pr}_1, g \circ \text{pr}_2): C \times D \rightarrow C' \times D'.$$

Exercise 1.2.19 (Exercise 1.12.3 (outside selection)). Show that there are natural isomorphisms

$$\text{id}_C \times \text{id}_D \cong \text{id}_{C \times D} \quad \text{and} \quad (f' \times g') \circ (f \times g) \cong (f' \circ f) \times (g' \circ g)$$

for functors $f: C \rightarrow C'$, $f': C' \rightarrow C''$, $g: D \rightarrow D'$ and $g': D' \rightarrow D''$.

Lemma 1.2.20 (Finite compatibility of products). *The comparison isomorphisms for pairing and product functoriality may be chosen so that the following properties hold through specified identifications:*

(1) *Pairing respects identity natural isomorphisms and vertical composites.*

(2) *The comparison*

$$(f, g) \circ \sigma \cong (f \circ \sigma, g \circ \sigma) \tag{1.3}$$

is natural in f , g , and σ , and is compatible with iterated precomposition.

(3) *The comparisons for identity and composition of product functors are natural in their inputs and compatible with whiskering.*

(4) *The displayed comparison functors carry identifications between natural isomorphisms to identifications between their images.*

Proof. For each $h, k: T \rightarrow C \times D$, use an inverse of the equivalence

$$(h \cong k) \xrightarrow{\sim} (\text{pr}_1 h \cong \text{pr}_1 k) \times (\text{pr}_2 h \cong \text{pr}_2 k).$$

Pairing the component comparisons gives the comparison for products. Its projections come equipped with identifications with those components. For identity and composition, the two projected pastings are identified by the whiskering laws. For iterated precomposition, they are identified by the primitive pentagon and triangle. Lift these identifications through the displayed equivalence using Corollary 1.2.12. Their action on identifications is the action of these functors on natural isomorphisms, obtained by whiskering. $\square$

Bibliography

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